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Poids et
Mesures

Internal Standard Reference Data for qNMR: Maleic Acid [ISRD-01]

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Édité par le BIPM
Pavillon de Breteuil
F-92312 Sèvres Cedex
FRANCE

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1. Introduction

Nuclear magnetic resonance (NMR) spectroscopy is well-established as the pre-eminent method for the qualitative structural analysis of organic molecules. The potential for its application for quantitative organic analysis was also recognized soon after NMR instruments became commercially available. [1] However it has only been recently, as spectrometer capabilities have achieved a level of accuracy and precision comparable to those attainable by chromatographic techniques, that this potential has been widely realized in practice. As a result quantitative NMR (qNMR) methods, particularly for the assignment of the purity of individual organic compounds, are now actively and extensively employed. [2], [3], [4], [5] As evidence of its increasing application in this role, a recent editorial in the *Journal of Medicinal Chemistry* [6] highlighted and recommended the general utility of "absolute quantitative ^{1}H NMR spectroscopy to determine the purity of biologically tested research compounds". Purity assignment by qNMR spectroscopy potentially also meets the metrological requirements for a primary ratio measurement procedure. [7] Validated qNMR methods [8], [9] are now being used, generally in combination with data obtained by orthogonal chromatographic techniques, to assign the purity of organic materials intended for use as Primary Reference Materials ISO 17511:2003 for individual organic analytes. [11], [12], [13] [14] The availability of properly characterized Primary Reference Materials is in turn an essential initial step in establishing the metrological traceability for measurement results for an organic analyte linked in a calibration hierarchy to a specific pure material. [15]

The assignment of the mass fraction purity of an organic analyte *A* by qNMR in solution using an internal standard *S* is based on measurement Equation (1) below.

$$w_{\mathrm{A}} = \frac{I_{\mathrm{A}}}{I_{\mathrm{S}}} * \frac{N_{\mathrm{S}}}{N_{\mathrm{A}}} * \frac{M_{\mathrm{A}}}{M_{\mathrm{S}}} * \frac{m_{\mathrm{S}}}{m_{\mathrm{A}}} * w_{\mathrm{S}} \qquad (1)$$

w_{A} is the mass fraction of the analyte in the material subject to assignment, w_{S} the independently established mass fraction content of the internal standard, I_{A} and I_{S} are the integrals of the quantified signals, N_{A} and N_{S} the number of ^{1}H nuclei contributing to each quantified signal, M_{A} and M_{S} the molar masses of the analyte and internal standard and m_{A} and m_{S} the masses of the samples of the analyte and internal standard used in preparation of the solution subject to the qNMR measurement.

In optimal cases where the data processing carried out by experienced operators the standard uncertainty for purity mass fraction assignments for non-problematic systems have been reported to reach the level of 1 mg g^{-1} on an absolute basis, equivalent to 0.1 % relative. [16], [17] Factors including, *inter alia*, the lineshape and multiplicity of the signals integrated, the extent of interferences from impurities present, the nature of the internal standard and solvent used, the magnetic field strength, hardware settings and performance characteristics of the spectrometer as well as the approach taken to transform the free induction decay (FID) signal generated by the NMR experiment and integrate the signals of the resulting frequency domain spectrum all contribute to the overall uncertainty of the assigned value. Evidently, regardless of the precision of a qNMR measurement, the overall (relative) measurement uncertainty of a qNMR assignment can never be smaller than that associated with the purity of the internal standard used to obtain the result.

The first goal of this document is to furnish general recommendations for the design of a qNMR experiment and for the undertaking of a quantitative ^{1}H NMR measurement using the internal standard approach to provide a measurement result traceable to the International System of Units (SI). [18] It should be noted that although these principles

should apply generally to quantitative measurement involving any NMR-active nuclei the recommendations in this document are only intended for assignments by ^{1}H qNMR.

The second goal is to describe a set of seven internal standard reference materials (ISRMs) which the Bureau International des Poids et Mesures (BIPM) in collaboration with the National Metrology Institute of Japan (NMIJ) propose constitute a "universal" set of higher-order, SI-traceable internal standards. Other groups have proposed specific compounds or sets of compounds suitable for use as qNMR internal standards. [12], [19], [20], [21] Although there is some commonality between the internal standards recommended in the current literature and our proposal, the focus of the earlier papers is primarily their suitability for application for purity assignments by qNMR rather than their utility as SI-traceable primary measurement standards.

At least one ISRM compound should be suitable for use for the assignment of a given organic compound soluble in a specified NMR solvent. The seven compounds constituting the "universal" ISRM set, together with an outline of their solubility and suitability for use in four representative deuterated NMR solvents, are described in Figure 1 below.

The third goal and the focus of this specific document is to provide guidance regarding the use and limitations of maleic acid as an ISRM for qNMR analysis.

In general, qNMR ISRM should consist of a stable crystalline solid which is:

- a high purity Certified Reference Material (CRM) produced and characterized by a National Metrology Institute (NMI) using methods other than qNMR or has been assigned by qNMR using an NMI CRM as the internal standard;
- predominantly one organic component ($w_S > 995$ mg g^{-1});
- value assigned with small standard uncertainty ($u(w_S) < 5$ mg g^{-1});
- providing unique NMR signals, either as singlet or simple multiplet resonances, having Lorentzian lineshape and narrow signal width; free of significant impurities interfering with areas to be integrated;
- inert in solution in NMR solvent;
- soluble in the chosen NMR solvent at a level in excess of 2 mg mL^{-1};
- readily handled for accurate mass determinations:
  - non-hygroscopic
  - non-volatile
  - not subject to electrostatic effects
- having a ratio of quantifiable protons to the molar mass of the ISRM sufficient to allow for practical gravimetric operations.

It is recognized that these characteristics constitute a "wishlist" rather than prescriptive requirements and that not all the materials constituting the ISRM suite described in this document meet all these specifications.

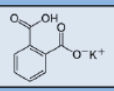
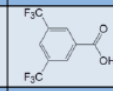
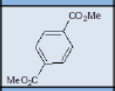
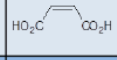
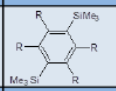
The solvents listed in Figure 1 are intended as representative of those with similar capabilities for solubilizing each ISRM rather than as a prescriptive set for use in qNMR. However these are the most readily available deuterated solvents and a majority of reported applications of solution qNMR have been undertaken using one of these solvents.

Compounds recommended as ISRMs for use with $CDCl_3$ as solvent (BTFMBA, DMTP, $DMSO_2$ and BTMSB) should be suitable for use in other chlorinated (CD_2Cl_2, $C_2D_2Cl_4$) or non-polar (benzene-d_6, toluene-d_8, THF-d_8, pyridine-d_5) solvents.

Likewise, compounds recommended as suitable ISRMs for use with DMSO-d_6 (BTFMBA, MA, $DMSO_2$ and DSS-d_6) are anticipated to be suitable for use in other polar organic solvents (acetonitrile-d_3, acetone-d_6, DMF-d_7).

At least three internal standards are applicable to each solvent class and provide quantification signals distributed across the standard ^1H chemical shift range.

Maleic acid (MA) is one of the seven proposed ISRMs. It is suitable for use as an internal standard for qNMR purity assignments of analytes soluble in D_2O and DMSO-d_6 and related solvents. The following section of this reference document and the attached annexes describes specific properties and applications of maleic acid.

ISRM ⇨	KHP	BTFMBA	DMTP	MA	$DMSO_2$	BTMSB	DSS-d_6
Structure							
δ (ppm)	8.3-7.0 (4H)	8.4-8.5 (2H) 8.2-8.4 (1H)	8.1 (4H) 3.9 (6H)	6.3 (2H)	3.0 (6H)	[7.5 (4H, R=H)] 0.2 (18H)	0.1 (9H)
Solvent ⇩	ISRM Solubility by Solvent ($mg.mL^{-1}$)						
D_2O	> 10	< 1	< 1	5-10	> 10	< 1	5-10
DMSO-d_6	< 2.5	> 10	< 2.5	> 10	5 - 10	< 5	5-10
CD_3OD	< 2.5	> 10	*	*	5 - 10	< 5	5-10
$CDCl_3$	< 1	5-10	> 10	< 1	> 10	5-10	< 1

* soluble but unsuitable due to (trans)esterification reaction with CD_3OD

Figure 1 — qNMR ISRM Suite [22]

Key

KHP Potassium hydrogen phthalate

BTFMBA 3,5-*bis*-Trifluromethylbenzoic acid

DMTP Dimethyl terephthalate

MA Maleic acid

$DMSO_2$ Dimethyl sulfone

BTMSB 1,4-*bis*-Trimethylsilylbenzene (R=H), BTMSB-d_4 (R = D), BTMSB-F_4 (R = F);

DSS-d_6 3-(Trimethylsilyl)-hexadeuteropropane-1-sulfonic acid [4,4-Dimethyl-4-silapentane-1-sulfonic acid-d_6]

D_2O Deuterium oxide

DMSO-d_6 Dimethyl sulfoxide-d_6 / Hexadeuterodimethyl sulfoxide

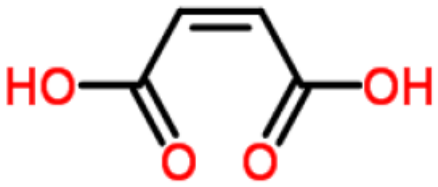
CD_3OD Methanol-d_4 / Tetradeuteromethanol

$CDCl_3$ Chloroform-*d* / Deuterochloroform

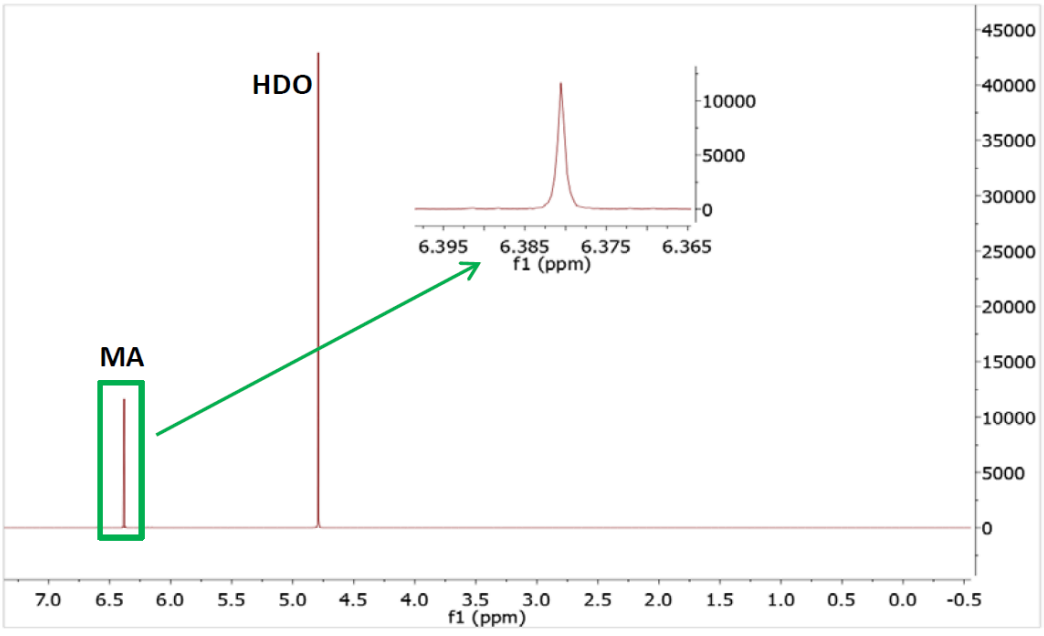
2. Properties of Maleic Acid

2.1. Physical Properties

Name	Maleic Acid
Structure	+



Synonym	(Z)-2-Butenedioic acid
CAS Registry Number	110-16-7
Molecular Formula	$C_4H_4O_4$
Molar Mass [23], [24]	116.072 g mol^{-1}, $u = 0.0025$ g mol^{-1}
Melting point [25]	138 °C
Density	1590 kg m^{-3} [25] 1526 kg m^{-3} [26]
Appearance	White crystalline powder
^{1}H NMR [27]	δ 11 (br. S, 2H) ; 6.29 (s, 2H)
^{13}C NMR	δ 166.5; 130.0



^{1}H NMR spectrum of MA in D_2O: JEOL ECS-400 spectrometer with Royal probe.

Example 400 MHz spectra of MA in D_2O and DMSO-d_6 are given in Annex A1.1.

2.2. Solvent Compatibility

NMR solvents suitable for use with MA are D_2O and DMSO-d_6. MA is soluble at levels in excess of 5 mg mL^{-1} in D_2O and in excess of 10 mg mL^{-1} in DMSO-d_6. [22]

Although MA is also soluble in CD_3OD, the formation in situ of mono- and di-esterification artefacts from reaction with the solvent preclude the use of MA for qNMR purity assignments in solution in CD_3OD or with other deuterated primary alcohols. [28]

2.3. Quantification signal

The two magnetically equivalent olefinic protons of maleic acid are observed as a singlet occurring at a chemical shift in the range 6.2 ppm – 6.4 ppm on the δ scale. The exact position of the resonance is a function of other factors including, but not limited to, the solvent, temperature, pH and the concentration of MA and other analytes in the solution. For optimal quantification results the homogeneity of the spectrometer magnetic field should be optimized such that the full width at half maximum (FWHM) of this signal is less than 1 Hz while the base of the resonance retains a suitable Lorentzian peak shape.

2.4. Impurities and artefact signals

Samples of maleic acid analysed in our laboratory have typically not presented evidence of the presence of significant levels (> 0.1 %) of related structure impurities in the material. There has also been little evidence of problems due to *in situ* isomerization of maleic acid to fumaric acid [(*E*)-2-butenedioic acid] in solution. However this could be problematic in aqueous solution at high pH or in the presence of an analyte containing a functional group capable of 1,4-conjugate addition to the olefinic bond. The presence or formation of fumaric acid is readily detected by the appearance of a singlet absorption at a chemical shift approximately 0.6 ppm downfield from the maleic acid olefin signal.

As noted, artefacts can also be formed by esterification with a primary alcohol. A set of NMR spectra illustrating the ongoing formation of mono- and di-d_3-methyl maleate in a solution of MA in CD_3OD analysed at 24 hour intervals is reported in Annex A1.3.

In practice the main interferences in a solution containing MA will come from the signals due to residual non-deuterated solvent. The chemical shifts of these signals are given in Table 1 below. Note that in the case of solutions in D_2O the signal due to residual HDO could potentially be attenuated if desired by the use of a (water) signal suppression pulse sequence, at the cost of introducing some additional non-linearity into the signal responses. [29]

2.5. Solvent recommendations and advisories

2.5.1. D_2O

D_2O is an excellent choice as solvent for use with MA. Rapid exchange of carboxyl protons with deuterons removes the potential for a broad signal and baseline interference due to the acidic hydrogens present in MA.

2.5.2. DMSO-d_6 and related solvents

Although relatively soluble in DMSO-d_6, use of MA in this solvent can be problematic due to potential interference from the signal due to the two acidic hydrogens. The extent of this interference is variable but can result in difficulties in the integration of signals and of interference with the spectrum baseline. If this interference is problematic it can be attenuated by the addition of D_2O as a co-solvent, at the cost of an additional or increased signal due to HDO.

Alternatively, multipoint baseline correction algorithms can be used to diminish or eliminate the extent of baseline distortion at the cost of introducing potential bias into the resultant integral quantifications. An example where this process was applied for a solution of MA and DMTP in DMSO-d_6 is given below in Annex A1.2.

2.5.3. Methanol-d_4 and related solvents

As discussed in Section 2.2 and Section 2.4 above, despite its solubility the use of MA in CD_3OD or other deuterated primary alcohols for high accuracy qNMR assignments is precluded due to the occurrence of an esterification reaction between the solvent and the MA carboxylic acid groups. [29]

2.5.4. Chloroform-*d* and related solvents

MA is not sufficiently soluble in $CDCl_3$ or other chlorinated or non-polar solvents.

Table 1. Solvent Parameters for MA

Solvent	qNMR signal - Singlet, ^2H (ppm) [a]	Integration range (ppm) [a]	T_1 (s) [a]	Residual Solvent (ppm)	Comments:
D_2O	6.4	6.1 – 6.7	6-7	**4.8** [b]	
DMSO-d_6	6.2	5.9 – 6.5	2.5-3.5	2.5	Potential for baseline interference from acidic protons of MA
CD_3OD		Not Suitable			*In situ* esterification
$CDCl_3$		Not Suitable			Insufficient solubility

(*a*) Indicative values only. The observed value in a specific qNMR solution will be a function of factors including concentration of MA and analyte, solution temperature, instrument, etc.

(*b*) Chemical shift of residual HDO signal is strongly pH dependent

3. Good Practice Guidance for SI-Traceable qNMR Measurement Results

3.1. Introduction

The first step in any purity assignment by qNMR should be the confirmation by qualitative NMR or other techniques of the identity of the analyte subject to purity assessment. In addition to confirming that the molar mass (M) and the number of nuclei (N) contributing to each signal subject to integration are appropriate, obtaining qualitative NMR spectra also provides a check for the occurrence and extent of any interfering signals in the sections of the NMR spectrum subject to integration.

Once the qualitative identity of the analyte has been appropriately established the input quantities that influence qNMR measurement results must be evaluated. These are identified from the measurement equation (Equation (1), Chapter 1, see p. 6). The purity of the internal standard used for the measurement, the source of traceability to the SI for the value assigned to the analyte, is established independently prior to the qNMR experiment.

The gravimetric procedure used for the preparation of the NMR solution has to be fully validated and fit for purpose, [30], [31] and the spectrometer performance, experimental parameters and the protocol for signal processing and integration must be optimized, [8], [9], [32] in order to produce a result for the ratio of the integral of the analyte and standard signals that accurately reflects the molar ratio of the hydrogen nuclei giving rise to the signals. [33] Only when these conditions are met can the assigned mass fraction purity of the analyte also be regarded as properly traceable to the SI. [11], [12], [34] Some general guidance for recommended practice for these critical steps is given in the following sections.

3.2. Internal standard

The internal standard used in qNMR should comply as far as possible with the criteria described in the Introduction regarding composition, physical characteristics, inertness, solubility, impurity profile and suitability for accurate gravimetry. In addition, in order to establish traceability of the result of the qNMR assignment to the SI, the material should comply with the requirements of a reference measurement standard, and in particular a reference material, as defined in the International Vocabulary of Metrology (VIM). [35]

To maintain SI-traceability the sources of the internal standard should be either a:

1. Certified Reference Material (CRM) characterized for its mass fraction purity and value assigned by an NMI;
2. CRM produced by a Reference Material Provider accredited to ISO 17034:2016 ISO 17034:2016 requirements;
3. High-purity material subject to a validated measurement procedure for purity assignment by qNMR using as an internal standard a CRM of type 1) or 2).

3.3. Gravimetry and Sample Size

The realization of accurate and precise qNMR measurements relies on the application of a properly implemented gravimetric procedure for the mass determinations of the internal standard and analyte. Recommended practice in this area in the specific context of qNMR sample preparation has been described in a recent publication. [30] Achieving an overall relative standard measurement uncertainty for the result of a qNMR assignment of 0.1 % requires the relative uncertainty associated with individual gravimetric operations typically to be less than 0.03 %. If the combined standard uncertainty of a mass determination is 3 μg, a level achievable with modern electronic microanalytical balances, this corresponds to a minimum sample size of 10 mg.

In addition to suitable control for each mass determination, if the receptacle used for the final solution preparation is not the same as that used for both mass determinations, the procedure for transfer of solids into the solution must address the assumption that the ratio of the gravimetric readings from the balance operations is equivalent to the ratio of the masses of each compound in the solution subject to the qNMR analysis.

For the examples reported in the Annex A1.2 below, gravimetric operations were undertaken using a balance associated with a measurement uncertainty estimate of 1.3 μg for individual mass determinations. In this case a minimum sample size of 4 mg achieves a relative uncertainty in individual gravimetric operations below 0.03 %. In addition to the measurement uncertainty of the gravimetric operations, high accuracy measurements require additional correction for sample buoyancy effects [31] and the $^{1}H/^{2}H$ isotope composition of the quantified signals. The value and associated uncertainty of the $^{1}H/^{2}H$ isotope composition of each quantification signal can be obtained using an on-line calculator application. [24]

As sample preparation for qNMR involves mass determinations in the milligram range using sensitive balances, the loss of even minute (almost invisible) quantities of powder during the gravimetric procedure will have a measurable influence on the balance reading and hence on the input quantities for the qNMR assignment. Environmental conditions for gravimetry and qNMR sample preparation should be controlled throughout the process, subject to minimum change and kept within the operating range recommended by the manufacturer. [37], [38] It is recommended that mass determinations be performed in an area where the relative humidity is maintained in the range 30 % to 70 %.

The accumulation of surface electrostatic charges is another potential source of bias for mass determinations, particularly for high-polarity, hygroscopic compounds. In these cases, treatment of the sample with an electrostatic charge remover or deioniser is advisable prior to the mass determination. Materials subject to qNMR analysis should be evaluated for their hygroscopicity, for example by measurement of the change in observed mass as a function of relative humidity using a dynamic sorption balance. This allows for assessment of the likely impact of variation in the relative humidity in the local environment on the results of gravimetric operations for a given compound. A minimum of two independent gravimetric sample preparations should be undertaken.

3.4. NMR spectrometer optimization

There is no specification of minimum NMR spectrometer field strength for purity measurements. Increasing the field strength enhances signal separation and sensitivity, both of which should increase the accuracy and precision of qNMR measurements. Careful

optimization of the lineshape (shimming) is critical in order to achieve reliable qNMR results. [39] A general guidance is to choose the simplest signal in the sample, often the residual solvent peak, and to optimize the instrument shimming until this signal is symmetrical with a FWHM below at least 1 Hz. Experience has shown that these lineshape requirements are more easily achieved using an inverse probe than a direct type. For lower field magnets (< 300 MHz), this requisite might not be attainable which impacts on the level of measurement uncertainty associated with the assigned value. In no case should a signal from a labile, exchangeable hydrogen or one subject to dynamic tautomeric exchange be used for quantitative measurements

Due to the relatively wide Lorentzian shape of NMR resonances the separation of the signals to be quantified from each other and from the remainder of the NMR signals in the spectrum should be considered carefully. Ideally there should be no interfering signals within a range one hundred times the FWHM on each side of each signal to be integrated.

3.5. NMR acquisition parameters

The basic experiment to perform quantitative NMR experiments uses a simple 1D pulse sequence designed to minimize differences in the integrated signal intensities due to differential rates of relaxation. For highest accuracy assignments, use of broadband heteronuclear decoupling should in general be avoided as it can lead to undesired nuclear Overhauser effects introducing a bias in the intensities of individual measured signals. However in the common case of ^{13}C-decoupling to remove satellite signals, the potential for bias is attenuated because of the low (1.1 %) natural abundance of the ^{13}C isotopomer even though the decoupling efficiency for individual ^{13}C satellite signals is variable. The potential for the introduction of additional bias due to ^{13}C-decoupling is negligibly small in most cases.

The basic sequence for a qNMR measurement consists of a “delay-pulse-acquire” experiment. There are critical parameters associated with each phase of the sequence in order to achieve a reliable, unbiased and quantitative signal response. Assuming the experiment starts from an equilibrium magnetization state, the first phase in the experiment is the pulse, which itself is preceded by a delay.

In the pulse phase, the two critical parameters for good qNMR measurement results are the pulse offset and pulse length (also called pulse width or tip angle). When a single “hard” pulse is applied to the bulk magnetization of each compound, off-resonance effects can occur if the frequency offset of the initial pulse is relatively far from that of the signals of interest. Ideally the pulse offset should be positioned as close as possible to the midpoint between the two signals to be quantified. This will not eliminate off-resonance effects but should result in cancelling out in both signals.

Regarding the pulse length, 90 ° pulses are recommended for quantitative analyses. A 30 ° pulse experiment, providing a signal response approximately half that of a 90 ° pulse, has the potential advantage of needing a significantly shorter relaxation time to re-establish equilibrium magnetization compared with a 90 ° pulse while requiring only twice as many transients to achieve an equivalent total **signal** response. However this potential advantage is offset by the need for four times as many transients as a 90 ° pulse to achieve the same **signal to noise** ratio. The accuracy of the results should not be impacted by the use of different pulse lengths but the acquisition time to achieve equivalent levels of precision will.

Additional parameters requiring optimization in the acquisition phase are the spectral window width, the acquisition time, the digital resolution and the relaxation delay time between acquisitions. The spectral window chosen will depend on the design and performance of the instrument used. The theoretical justification for the use of a large spectral window is that oversampling the FID will produce noise filtering. However, the efficiency of digital filters varies by instrument and the appropriate spectral window should be evaluated on a case-by-case basis.

The acquisition time should be at least 2.5 s to avoid truncation of the signals and to allow good digitisation of the spectrum. The ideal acquisition time is the smallest time for which no truncation is observed. Use of longer acquisition times than necessary primarily results in addition of noise to the spectrum. The digital resolution should not exceed 0.4 Hz pt^{-1} in order to have accurately defined signals that will give accurate area measurements and suitable precision at typical sampling rates.

The relaxation delay between pulses in particular has to be carefully established for each sample mixture. To determine the optimum repetition time for a given qNMR measurement it is critical to determine the longest T_1 time constant of the signals to be quantified. This document presents some observed values measured for maleic acid in different solvents at the concentration and under the specific instrumental conditions used, but these should be regarded as indicative only, and in any event they are not the determining factor in cases where the T_1 of the analyte quantification signal is longer.

As the T_1 constant arises from a process of spin-lattice relaxation, its values are strongly dependent on the composition of the solution being measured and it should be determined for each signal to be quantified in each mixture on a case-by-case basis. The most commonly used method to determine the T_1 constant is the inversion-recovery sequence, which is generally available in the factory programmed pulse sequences installed with any NMR. The application of the inversion recovery experiment requires knowledge of the optimized 90 ° pulse, which should also be determined for each mixture under investigation. The 90 ° pulse is used for both the T_1 determination and the quantitative measurements.

The repetition time between pulses should correspond to the full loop time in the pulse sequence and not simply the relaxation delay. Since most of the time intervals involved in NMR measurement are negligible relatively to the T_1 values, the repetition time (RT) can be estimated as the sum of acquisition time (AQ) and relaxation delay (RD), where the RD is a multiple T_1. After a 90 ° pulse, if available instrument time permits, 10 times T_1 of the signal with the longest relaxation time will lead to the recovery of > 99.995 % of the magnetization for all quantified signals. In cases where the T_1 of the quantified signals are similar in magnitude, a shorter relaxation delay may be sufficient for equivalent (even if incomplete) magnetization re-equilibration.

Thus the pulse RT is given by:

$$\mathrm{RT} = \mathrm{RD} + \mathrm{AQ} = n * T_1 \qquad (2)$$

$$(n = 10\text{–}15)$$

The number of transients (scans) should be determined according to the concentration of the sample, the nature of the signals and the available instrument time. To achieve small uncertainty a signal to noise (S/N) ratio of at least 1000 should be achieved for each signal subject to quantification. Smaller S/N values can still lead to acceptable results, but the reported measurement uncertainties increase as the S/N ratio decreases.

Table 2. Recommended NMR Parameters for quantitative measurements.

Parameter	Recommended Value	Explanation/Comments
Shimming	FWHM of lineshape signal (eg $CHCl_3$/ acetone-d_6) < 1 Hz	Optimization of field homogeneity is critical for uniform response over typical chemical shift range
Pulse Width	90 °	Should not change the quality of the results, but the use of a 90 ° pulse with adequate recovery time leads to a smaller total acquisition time for a target S/N ratio.
Pulse Offset	Midpoint between signals	Theoretically makes off resonance effects equivalent
Repetition Time	$10-15 \times T_1$	After 90 ° pulse, a delay of 10 T_1 of the signal with the longest relaxation time necessary for recovery of > 99.995 % of magnetization for all quantified signals.
Number of Transients (scans)	As needed for adequate signal to noise ratio	Evaluate on a case by case basis. Minimum requirement is S/N > 1000 for each signal quantified
Spectral Window	> 20 ppm	The use of a wide spectral window for data recording (oversampling) has been reported to yield better results in some instruments because of the noise filtering it produces in the quadrature detection scheme. This is instrument dependent and should be evaluated.
Acquisition Time	> 2.5 s	The correct acquisition time is essential to give the best digital resolution for good quantitative results. If too short, lower digital resolution and truncated signals result. If too long excessive noise is introduced. A minimum of 2.5 s is a useful starting point and 4 s has been found to be suitable for many applications.
Digital resolution	< 0.4 Hz pt^{-1}	The digital resolution is the reciprocal of the acquisition time. Suitable signal shape sensitivity requires not less than 0.4 Hz pt^{-1}.
Signal Integral Ratio	1:1	The preference are sample sizes such that the integral ratio for the quantification signals is close to equivalent. However in practice this ratio can vary within the range 10:1 to 1:10 provided the S/N ratio of the lower intensity peak is > 1000.

Good practice for performing quantitative experiments is to prepare, in addition to the sample mixtures, one sample consisting of a solvent blank, one with the analyte only and one with the internal standard only in the same solvent. These additional NMR spectra should be acquired prior to the preparation of sample mixtures to check the suitability of the proposed mixture in

terms of the absence of interferences from one compound (or impurities present in it) in the other. Other NMR techniques such as 2D HSQC or COSY may be applied to demonstrate the uniqueness of the signals used for quantification and the absence of overlapping contributions from impurities while aware that the sensitivity of such techniques is low and the absence of observable interferences does not guarantee a signal free of such interferences.

Each analyte/IS mixture should be measured at least three times in the NMR system. Independent measurements for a particular sample mixture should be non-continuous, where the tube is removed and the measurement process (tuning, locking, shimming) is repeated each time for each sample. To avoid potential unwanted contributions due to spinning sidebands, it is recommended to undertake the measurement using sample spinning disabled. This presumes a high degree of field homogeneity has been achieved.

3.6. NMR signal integration

The integration range should extend on each side at least seventy six times the FWHM of the signal being measured in order to integrate in excess of 99.9 % of the signal. The estimation of signal width should be done for the outer signals if a multiplet signal is subject to integration. An alternative method that generally produces acceptable results is to use a range extending 30 Hz beyond the furthest ^{13}C satellites as the start and end points for the integration ranges, as this generally exceeds the above described width. It is important to apply a suitable algorithm for the baseline correction and check its validity by analysing standard samples. Practical experience has shown that manual baseline assignment currently works best when high accuracy qNMR results are required. [32], [39] A window function can be applied as a final data treatment parameter to enhance the S/N ratio. [9] To avoid line broadening effects, an exponential multiplication factor not greater than 0.3 Hz should be used. The window function in use at the BIPM with the JEOL-ECS 400 was typically no greater than 0.05 Hz — 0.10 Hz and in some cases it was not used at all.

3.7. Measurement uncertainty

Evaluation of the measurement equation previously presented (**Equation (1)**) allows for identification of individual factors potentially influencing the input quantities for the measurement uncertainty as shown in the diagram in **Figure 2.**

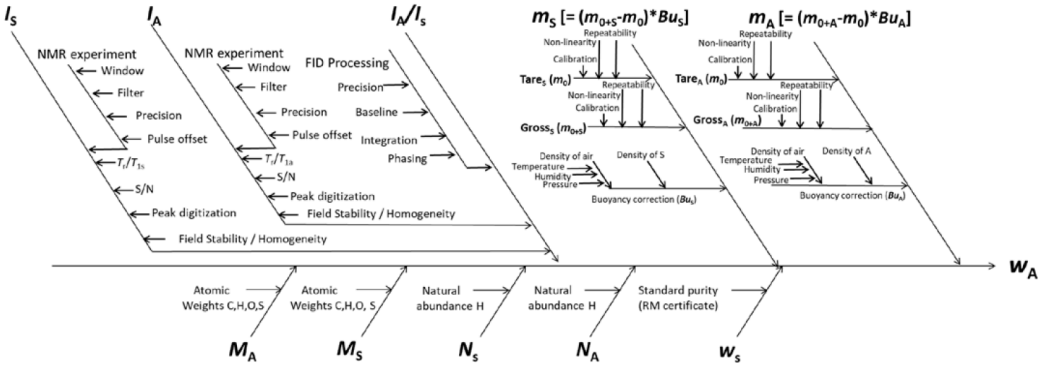


Figure 2 — Ishikawa diagram for input quantities considered for the measurement uncertainty estimation by qNMR

The observed repeatability of the integral area ratios, which incorporates contributions from the input factors for excitation, population, detection efficiency and data processing, is amenable to a type A statistical evaluation. [12], [32], [40] Since these measurements should come from at least two independent solutions each containing different sample masses, the area ratios will vary on a sample-by-sample basis.

The measurement uncertainty of the value obtained for each preparation can be evaluated separately and the individual purity results for each sample combined statistically. Another approach is to pool the purity values from the replicate results for the separate samples. Analysis of this combined data by ANOVA produces an assigned value and provides an estimate of the intermediate precision of the overall process. It also identifies if additional variance contributions from sample preparation and signal processing contribute significantly in addition to that arising from the method repeatability.

The final assigned value will be similar regardless of the approach used, although the contribution to the measurement uncertainty of the result may differ.

The standard uncertainties for the other major input quantities are type B estimates and are straightforward to evaluate. Molar masses and the $^{1}H/^{2}H$ isotope distribution of the quantification signals, with their associated uncertainties, were calculated based on the values for atomic weights and hydrogen isotope distribution in the 2016 revision of the IUPAC Technical report of the Atomic weights of the elements, [23], [24] the uncertainties of individual gravimetric operations are based on balance performance characteristics corrected for buoyancy effects [16] and the uncertainty of the purity of the internal standard is assigned by the material provider.

Other approaches to the evaluation of measurement uncertainty for qNMR and the combination of results from qNMR with orthogonal techniques for purity evaluation have also been reported. [8], [11], [12], [33], [41] Example measurement uncertainty budgets for qNMR analysis are provided in Annex A1.2.

4. Acknowledgements

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All NMR studies were carried out by the co-authors of this document in the course of secondments at the BIPM. The support of the parent institution of each scientist in making them available for secondment to the BIPM is gratefully acknowledged.

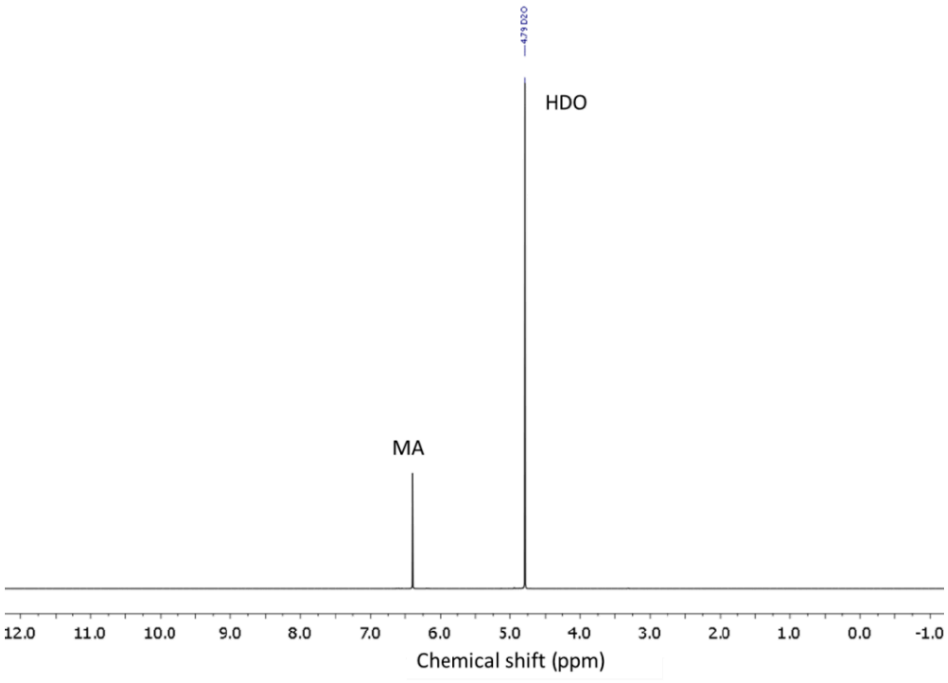
Dr Bruno Garrido wishes to acknowledge funding for his secondment from the Brazilian Ministry of Education under the Coordination for the Improvement of Higher Education Personnel (CAPES) post-doctoral scholarship programme (process: 99999.007374/2015-01).

DISCLAIMER: Commercial NMR instruments, software and materials are identified in this document in order to describe some procedures. This does not imply a recommendation or endorsement by the BIPM nor does it imply than any of the instruments, equipment and materials identified are necessarily the best available for the purpose.

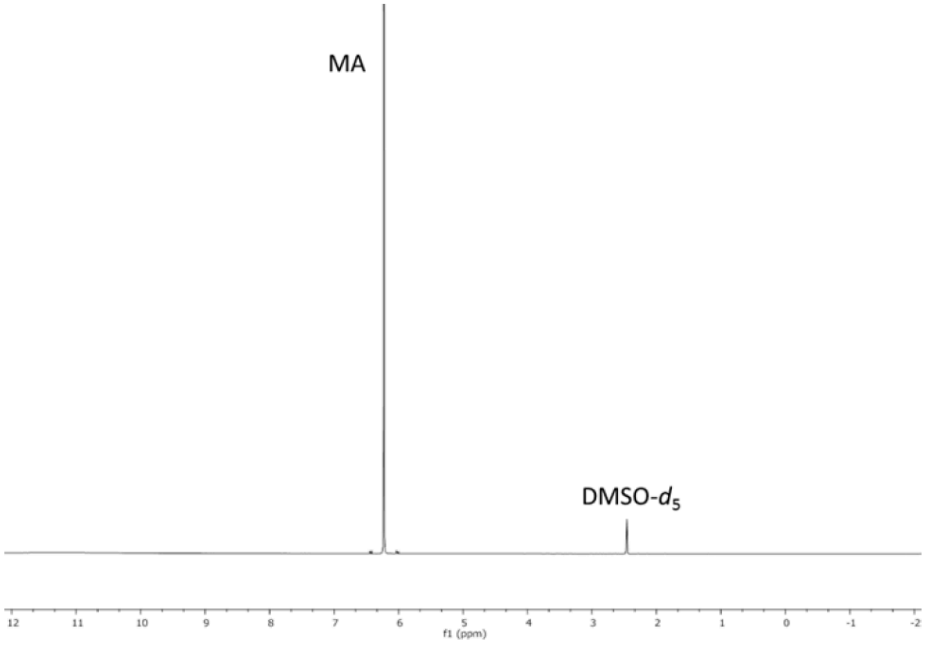
Annex 1. Annexes

A1.1. Solution NMR Spectra of Maleic Acid

A1.1.1. MA in D_2O



A1.1.2. MA in DMSO-d_6



A1.2. qNMR using MA as internal standard

Two examples are provided of the value assignment by qNMR of the mass fraction content of organic compounds using MA as the ISRM. In the first example MA was used in a solution in D_2O with $DMSO_2$ as analyte. [22] In the second example DMSO-d_6 was the solvent with DMTP as the analyte. [42]

These are intended as "best case" illustrations and should not be regarded as representative of the uncertainty budget achievable when quantifying more complex resonance signals or with more structurally complex compounds. The signals for quantification in these examples are clearly separated from each other, have narrow, well-resolved signal shape and there is no significant interference from impurities or solvent. As a result the uncertainty contribution due to the reproducibility of the signal integration is smaller (and the relative uncertainty contribution due to the uncertainty associated with gravimetry and the purity of the internal standard correspondingly greater) than would be anticipated for more typical routine applications.

Regular shimming was used to maximize the homogeneity of the instrument field. Gravimetric determinations were carried out using a microbalance with readability of 0.1 μg and a measurement uncertainty for an individual net mass of less than 100 mg of 1.3 μg.

The MA was obtained from a commercial provider and used without additional treatment. The purity was assigned in separate qNMR experiments in solution in D_2O using a high-purity CRM for KHP (either NIST SRM 84L or NMIJ CRM 3001b) as the internal standard. The mass fraction content of the MA by our internal measurement was 999.7 ± 0.6 mg g^{-1}, consistent with the purity value for the material reported by the material provider.

The $DMSO_2$ and DMTP used as analytes and all deuterated solvents were purchased from commercial suppliers and used without further treatment or purification. Commercial borosilicate glass NMR tubes with 5 mm internal diameter rated for use in 500 MHz spectrometers were used for all measurements.

A1.2.1. MA (IS) & $DMSO_2$ (Analyte) in D_2O

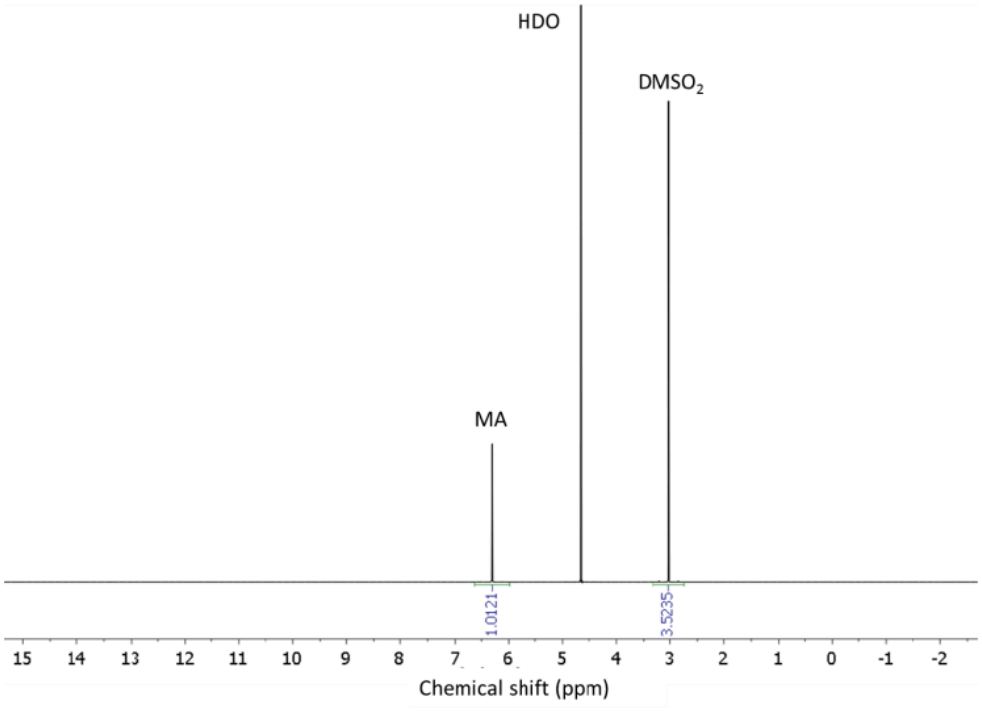


Figure A1.1 — ^1H NMR spectrum of MA + $DMSO_2$ in D_2O.

The optimized gravimetric and NMR parameters for the qNMR assignment using a JEOL ECS-400 spectrometer equipped with a Royal probe are given in Table 2. The sample was made up in solution in approximately 1 mL of D_2O and 800 µL was transferred into the NMR tube for analysis.

Table A1.1. NMR experiment parameters for $DMSO_2$ purity assignment using MA in D_2O

Parameter	Value
MA Sample size (mg)	2.3 – 3.7
$DMSO_2$ Sample size (mg)	3.5 – 4.9
Number of Transients	32
Receiver gain	Automatic
Acquisition time (s)	4
Relaxation delay (s)	65
Pulse offset (ppm)	4.8
Spectral width (ppm)	400

Data points	639652
Temperature (K)	298
Spinning	Off
Integral ratio (MA:$DMSO_2$)	0.13 – 0.29 [(a)]

(*a*) integral ratio reported for information only — not necessarily "optimal" value

A baseline correction window of one hundred times the FWHM was used for each integrated signal. The integration range covered eighty times the FWHM. Four independent sample mixtures were prepared and each sample was measured four times. The measurement uncertainty budget for one of the samples is reproduced in Table A1.2. The integral ratio is the mean of the four replicate values obtained for this sample. The standard uncertainty of the ratio is the standard deviation of the mean. The other uncertainty components are Type B estimations. The relative contribution of each component to the uncertainty of the combined result for this sample is displayed in Figure A1.2. The mass fraction content of $DMSO_2$ assigned for this sample was 997.3 ± 1.5 mg g^{-1}.

Table A1.2. Uncertainty budget for $DMSO_2$ purity by qNMR using MA as ISRM in D_2O.

Uncertainty source	Value	Uncertainty Evaluation Type	Standard Uncertainty	Sensitivity coefficient	Relative Uncertainty
I_A	97943	-	-	-	-
I_S	17834	-	-	-	-
Integral A/ Integral S	5.4919	A	0.00096	0.181587363	1.95E-04
Analyte signal ^1H Nuclei	5.9934	B	0.0003	-0.166397551	4.99E-05
IS signal ^1H Nuclei	1.9994	B	0.0002	0.498783696	9.98E-05
Analyte Molar Mass (g mol^{-1})	94.136	B	0.005	0.010594109	5.22E-05
IS Molar Mass (g mol^{-1})	116.0724	B	0.0025	-0.00859194	2.15E-05
Analyte Sample Mass (mg)	3.5063	B	0.00124	-0.284427197	3.53E-04
IS Sample Mass (mg)	2.3545	B	0.00124	0.423566396	5.26E-04

IS Purity (g g^{-1})	0.9995	B	0.0003	0.997785973	2.99E-04
Assigned value (g g^{-1})	0.9973		0.00073		7.32E-04
				Combined Uncertainty	0.000730
				ν_{eff}	784
Analyte mass fraction (g g^{-1}):	0.9973	± 0.0015		k	2
Analyte purity (% mass):	99.7	± 0.2		Expanded Uncertainty	0.00146

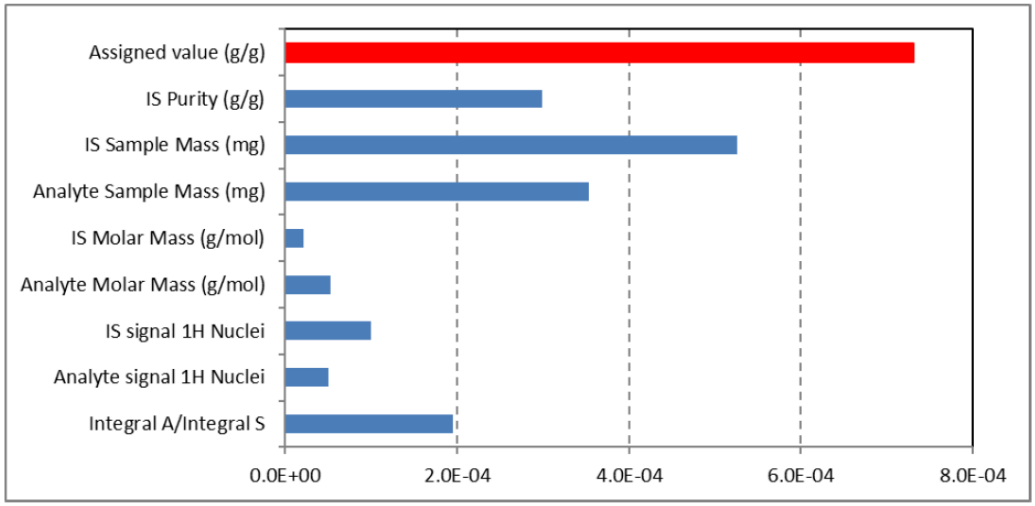


Figure A1.2 — Relative uncertainty components (in blue) for the uncertainty in the assigned purity value (in red) for $DMSO_2$ using MA as ISRM in D_2O.

A1.2.2. MA (IS) and DMTP (Analyte) in DMSO-d_6

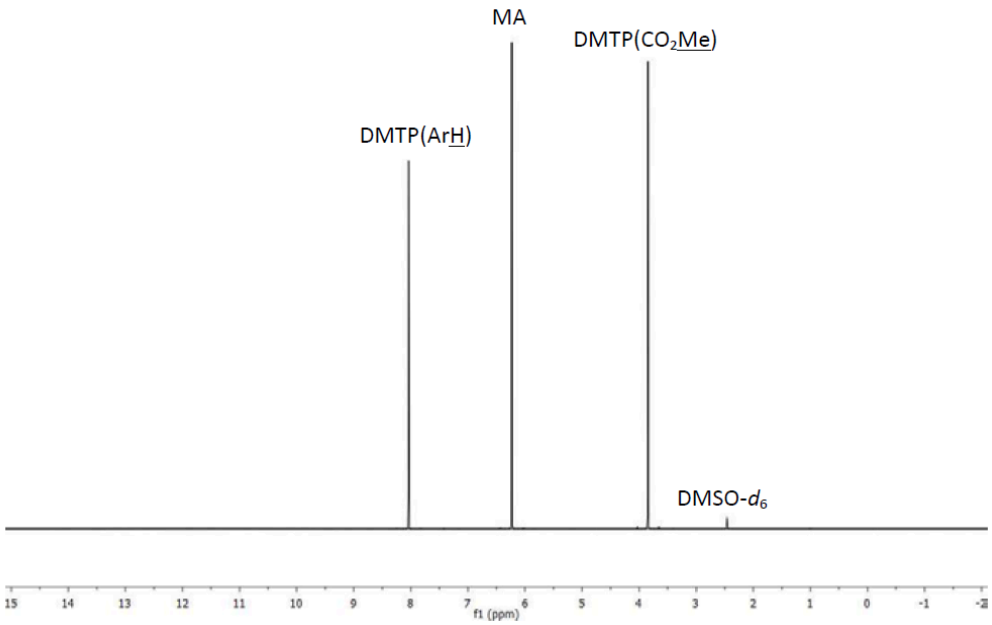


Figure A1.3 — ^{1}H NMR of MA + DMTP in DMSO-d_6 after baseline correction.

This is an example of a case in which D_2O was not a suitable solvent since DMTP is not water soluble. Multipoint baseline correction procedures were required to achieve acceptable baseline stability.

The experimental NMR parameters used for the measurement are given in Table A1.3.

Table A1.3. NMR experiment parameters for DMTP assignment using MA in DMSO-d_6.

Parameter	Value
MA Sample size (mg)	20 – 30
$DMSO_2$ Sample size (mg)	32 – 46
Number of Transients	32
Receiver gain	34
Acquisition time (s)	4
Relaxation delay (s)	50
Pulse offset (ppm)	7.0
Spectral width (ppm)	400

Data points	39979
Temperature (K)	298
Spinning	Off
Integral ratio (MA:$DMSO_2$)	0.8 – 1.3

Baseline correction was performed over the whole spectral width using a multipoint baseline correction algorithm. The integration range start and end points were placed 30 Hz beyond the ^{13}C satellite signals. Results from five independent sample mixtures each measured six times were obtained. The measurement uncertainty budget for one of the samples from the results for six replicate determinations is reproduced below in Table A1.4. The relative contribution of each component to the uncertainty of the result obtained for this sample is displayed in Figure A1.4. The mass fraction content of DMTP was 999.5 ± 0.8 mg g^{-1}.

Table A1.4. Uncertainty budget for DMTP purity by qNMR using MA in DMSO-d_6.

Uncertainty source	Value	Uncertainty Evaluation Type	Standard Uncertainty	Sensitivity coefficient	Relative Uncertainty
I_A	0.7501	-	-	-	-
I_S	1.0000	-	-	-	-
Integral A/ Integral S	0.7501	A	0.00012	1.3324	1.65E-04
Analyte signal ^1H Nuclei	3.9968	B	0.0002	-0.2501	5.00E-05
IS signal ^1H Nuclei	1.9994	B	0.0002	0.4999	1.00E-04
Analyte Molar Mass (g mol^{-1})	194.184	B	0.006	0.0051	3.09E-05
IS Molar Mass (g mol^{-1})	116.072	B	0.0025	-0.0086	2.15E-05
Analyte Sample Mass (mg)	20.3108	B	0.00337	-0.0492	1.66E-04
IS Sample Mass (mg)	32.3521	B	0.00337	0.0309	1.04E-04
IS Purity (g g^{-1})	0.9995	B	0.0003	1.0000	3.00E-04
Assigned value (g g^{-1})	0.9995		0.0004	1	3.96E-04

			Combined Uncertainty	0.0004
			ν_{eff}	131
Analyte mass fraction (g g^{-1}):	0.9995	± 0.0008	k	1.97824
Analyte purity (% mass):	99.95	± 0.08	Expanded Uncertainty	0.0008

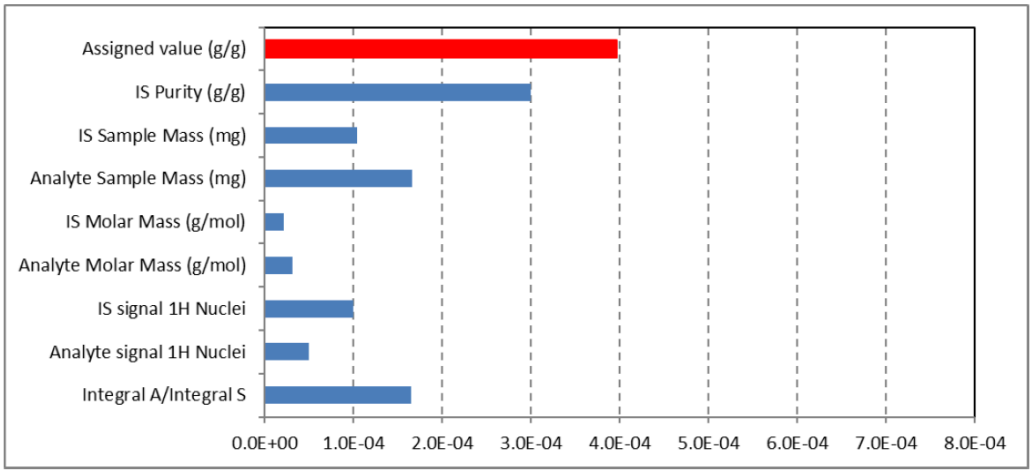


Figure A1.4 — Relative uncertainty components (in blue) for the uncertainty in the assigned purity value (in red) for DMTP when using MA as ISRM in DMSO-d_6.

Despite the drawback in cases such of the need for multipoint baseline correction in the whole spectrum, the result obtained for the purity assignment of DMTP agreed within its associated uncertainty with values obtained by qNMR analyses using other IS/solvent combinations, including those which did not require such extensive baseline correction.

A1.3. Time course for esterification of MA in solution in CD_3OD

Figure A1.5 illustrates the formation over time of esterification product, consistent with formation of a mixture of di- and mono-d_3-methyl maleate, after a sample of MA is taken up in solution in CD_3OD. [18] A second signal for the olefinic protons due to the esterification product appears downfield from the corresponding signal in unmodified MA. The signal area of the esterification product as a percentage of the parent MA signal is also shown.

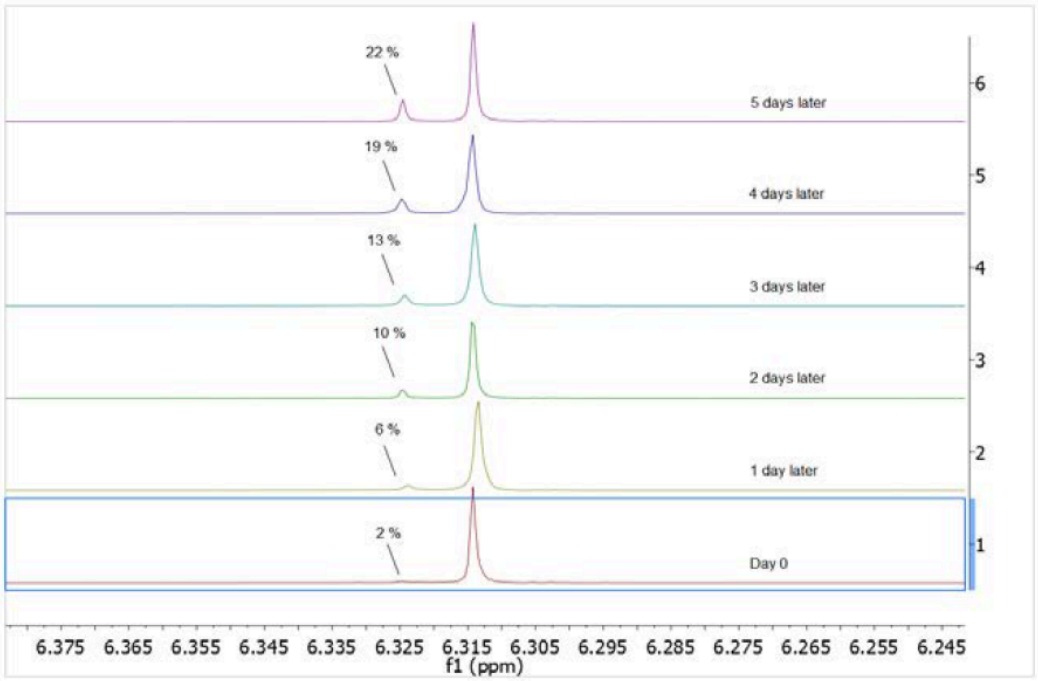


Figure A1.5 — NMR spectra of MA solution in CD_3OD.

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Authors: Steven Westwood (BIPM), Norbert Stoppacher (BIPM), Bruno Garrido (INMETRO, Brazil), Ting Huang (NIM, China), Takeshi Saito (NMIJ, Japan), Ilker Un (TUBITAK UME, Turkey), Taichi Yamazaki (NMIJ, Japan), and Wei Zhang (NIM, China)


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